

Arthur Cancellieri Pires

Data Scientist / PhD Candidate

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Public project: Machine Anomaly Detection + Alarms

Professional Experience

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- Railway Laboratory (Lafer) – partnership with VALE** On-site
TECHNICAL LEAD IN APPLIED AI *January 2022 to Present*
 - Led a team of 12 delivering a predictive maintenance program for VALE, one of the world's largest mining and logistics companies, converting stakeholder needs into shipped ML systems.
 - Cut equipment failure detection latency from 3 months to real time at 91% precision using deep learning models and automated data quality checks on multivariate sensor streams.
 - Flagged critical defects weeks before scheduled maintenance, including cases absent from the maintenance database, with unsupervised Transformer autoencoders on time series data.**Skills:** Python + libraries (PyTorch, PyTorch Lightning, scikit-learn) · Deep Learning (Transformers, Autoencoders) · Anomaly Detection · Time Series · Deployment (FastAPI, Docker)
 - Railway Laboratory (Lafer) – partnership with VALE** On-site
RESEARCH DATA SCIENTIST *June 2020 to January 2022*
 - Built a daily ETL pipeline processing 20,000 to 40,000 sensor records per asset per day, integrating multiple external sources into one analytical dataset that powered all downstream models.
 - Reduced reliance on costly specialized inspections by predicting asset condition at $R^2 \approx 0.98$ with deep learning models on sensor data.**Skills:** Python + libraries (scikit-learn, PyTorch) · SQL, PostgreSQL · ETL · Deep Learning · Signal Processing
 - Tribology and Railway Dynamics Laboratory (LabTDF) – partnership with VALE** On-site
INTERN IN MACHINE LEARNING AND OPTIMIZATION *Jan 2019 to June 2020*
 - Estimated hard to measure contact forces at $R^2 \approx 0.97$ from onboard sensors validated against simulation, making the company's predictive maintenance program viable.
 - Lowered component wear by 20% and improved a key safety ratio by roughly 35% by optimizing wear profiles with genetic algorithms on field data.
 - Drove a maintenance threshold change adopted into operations, projected to extend asset service life by 29% to 50%.**Skills:** Python, MATLAB · Machine Learning (scikit-learn, Optuna) · Genetic Algorithms / Optimization · Multibody Dynamics (Universal Mechanism, Simpack)

Education

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- University of Campinas (UNICAMP)** Campinas, Brazil
PH.D. IN MECHANICAL ENGINEERING *Expected: August 2026*
 - Thesis:** A methodology for detecting track defects from instrumented railway vehicle data.
 - University of Campinas (UNICAMP)** Campinas, Brazil
M.SC. IN MECHANICAL ENGINEERING *Jan 2022*
 - Thesis:** Measuring railway track irregularities from instrumented railway vehicle data using machine learning techniques.
 - Federal University of Espírito Santo (UFES)** Vitória, Brazil
B.SC. IN MECHANICAL ENGINEERING *May 2020*
 - Thesis:** Indirect identification of wheel–rail contact forces of an instrumented heavy haul railway vehicle using machine learning.

Languages

Portuguese (Native), English (Fluent), Italian (Basic)

Publications & Conferences

SELECTED PUBLICATIONS

- 2024 **Measuring vertical track irregularities from instrumented heavy haul railway vehicle data using machine learning**, Engineering Applications of Artificial Intelligence
- 2023 **Influence of wheel tread wear on Rolling Contact Fatigue and on the dynamics of railway vehicles**, Wear
- 2022 **Nonintrusive Operator Inference for Chaotic Systems**, IEEE Transactions on Artificial Intelligence
- 2022 **Generation of realistic artificial track irregularities for multibody simulation using measured geometric data: from mid-chord to space-curve**, In Proceedings of the Fifth International Conference on Railway Technology
- 2021 **Indirect identification of wheel–rail contact forces of an instrumented heavy haul railway vehicle using machine learning**, Mechanical Systems and Signal Processing
- 2021 **The effect of railway wheel wear on reprofiling and service life**, Wear

CONFERENCE PRESENTATIONS

- 2026 International Conference on Railway Technology (Railways 2026), Aug. 23–26 (accepted) *Budapest, Hungary*
- 2025 International Heavy Haul Association (IHHA) Conference, Nov. 17–21 *Colorado Springs, USA*
- 2023 International Heavy Haul Association (IHHA) Conference, Aug. 27–31 *Rio de Janeiro, Brazil*